

6.9 Thermal characteristics

The operating junction temperature T_J must never exceed the maximum given in [Table 24: General operating conditions](#).

The maximum junction temperature in °C that the device can reach if respecting the operating conditions, is:

$$T_J(\text{max}) = T_A(\text{max}) + P_D(\text{max}) \times \Theta_{JA}$$

where:

- $T_A(\text{max})$ is the maximum operating ambient temperature in °C,
- Θ_{JA} is the package junction-to-ambient thermal resistance, in °C/W,
- $P_D = P_{\text{INT}} + P_{\text{I/O}}$.
 - P_{INT} is power dissipation contribution from product of I_{DD} and V_{DD}
 - $P_{\text{I/O}}$ is power dissipation contribution from output ports where
 $P_{\text{I/O}} = \sum (V_{\text{OL}} \times I_{\text{OL}}) + \sum ((V_{\text{DDIO1}} - V_{\text{OH}}) \times I_{\text{OH}})$, taking into account the actual V_{OL} / I_{OL} and V_{OH} / I_{OH} of the I/Os at low and high level in the application.

Table 76. Thermal resistance

Symbol	Parameter	Package ⁽¹⁾	Value	Unit
Θ_{JA}	Thermal resistance junction-ambient	WLCSP15	126.3	°C/W
		TSSOP20	84.5	
		UFQFPN28	64.2	
		UFQFPN32	44.3	
		LQFP32	54	
		UFQFPN48	33.9	
		LQFP48	54	
Θ_{JB}	Thermal resistance junction-board	WLCSP15	96.5	°C/W
		TSSOP20	58.3	
		UFQFPN28	28.8	
		UFQFPN32	26.3	
		LQFP32	31.1	
		UFQFPN48	18.3	
		LQFP48	31.1	
Θ_{JC}	Thermal resistance junction-case	WLCSP15	10.4	°C/W
		TSSOP20	32.2	
		UFQFPN28	31.0	
		UFQFPN32	32.1	
		LQFP32	18.6	
		UFQFPN48	14.7	
		LQFP48	18.6	

1. Refer to [Section 6: Package information](#) for package dimensions